Huawei Access Node JAY-ACC-77 Routing Protocol Flap Resolution SOP

Node: JAY-ACC-77 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8956

Description

The failure of the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on the Huawei Access Node JAY-ACC-77 at the Jayanagar Metro Site can be attributed to the complex interplay of routing protocol mechanics, specifically the interaction between the interior gateway protocols (IGPs) and the exterior gateway protocol (EGP) BGP. The OSPF and ISIS LSP flooding can be triggered by various factors such as changes in the network topology, BGP route churn caused by updates to the BGP routing table, or BGP neighbor state changes. The Huawei Access Node, being an integral part of the network infrastructure, plays a critical role in facilitating the exchange of routing information between the different network segments. In this context, the Routing Protocol Flap on the Huawei Access Node JAY-ACC-77 can be caused by a range of factors including misconfiguration, hardware or software issues, or even environmental factors affecting the network equipment. To effectively troubleshoot and resolve the issue, it is essential to understand the underlying routing protocol mechanics and the specific Huawei CLI commands that can be used to diagnose and rectify the problem.

Pre-requisite Checks

- check: Verify that the remote hands are dispatched to the Jayanagar Metro Site to perform physical checks on the Huawei Access Node JAY-ACC-77., status: Completed

- check: Check the status of the redundant links connected to the Huawei Access Node JAY-ACC-77 to ensure that they are functioning correctly., status: Completed

Potential Root Causes

Cause	Probability
BGP Route Churn due to network topology changes	High
OSPF Adjacency Loss due to incorrect OSPF configuration	Medium
ISIS LSP Flooding due to hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table all, Expected Output: BGP routing table information, Description: This command is used to display the BGP routing table information, including the BGP routes, next hops, and AS paths. This can help identify any BGP route churn issues.

Command: display ospf interface brief, Expected Output: OSPF interface brief information, Description: This command is used to display the brief information about the OSPF interfaces, including the interface
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ID, state, and priority. This can help identify any OSPF adjacency
loss issues.

Command: display isis lsp flooding, Expected Output: ISIS LSP flooding
information, Description: This command is used to display the ISIS LSP
flooding information, including the LSP ID, flooding reason, and
flooding status. This can help identify any ISIS LSP flooding issues.
```

Resolution Steps

Step 1: Step: Verify BGP route churn issues, Command: display bgp routing-table all, Description: Verify the BGP routing table information to identify any BGP route churn issues. Check for any inconsistencies in the BGP routes, next hops, and AS paths.

Step 2: Step: Verify OSPF adjacency loss issues, Command: display ospf interface brief, Description: Verify the OSPF interface brief information to identify any OSPF adjacency loss issues. Check for any inconsistencies in the interface ID, state, and priority.

Step 3: Step: Verify ISIS LSP flooding issues, Command: display isis lsp flooding, Description: Verify the ISIS LSP flooding information to identify any ISIS LSP flooding issues. Check for any inconsistencies in the LSP ID, flooding reason, and flooding status.

Step 4: Step: Clear BGP route churn, Command: clear bgp *, Description: Clear the BGP routing table to resolve any BGP route churn issues.

Step 5: Step: Reset OSPF adjacency, Command: reset ospf adjacency, Description: Reset the OSPF adjacency to resolve any OSPF adjacency loss issues.

Step 6: Step: Clear ISIS LSP flooding, Command: clear isis lsp flooding, Description: Clear the ISIS LSP flooding information to resolve any ISIS LSP flooding issues.

Post-Check Verification

```
Command: display bgp routing-table all, Expected Output: BGP routing
table information, Description: Verify the BGP routing table
information to ensure that the BGP route churn issues have been
resolved.

Command: display ospf interface brief, Expected Output: OSPF interface
brief information, Description: Verify the OSPF interface brief
information to ensure that the OSPF adjacency loss issues have been
resolved.

Command: display isis lsp flooding, Expected Output: ISIS LSP flooding
information, Description: Verify the ISIS LSP flooding information to
ensure that the ISIS LSP flooding issues have been resolved.
```

Roll-back Procedures

- Step: Restore BGP routing table, Command: restore bgp routing-table, Description: Restore the BGP routing table to its previous state in case the fix fails.

- Step: Restore OSPF adjacency, Command: restore ospf adjacency, Description: Restore the OSPF adjacency to its previous state in case the fix fails.

- Step: Restore ISIS LSP flooding, Command: restore isis lsp flooding, Description: Restore the ISIS LSP flooding information to its previous state in case the fix fails.